



Intro to BAM GNPL Model V1.43.1

Bloomberg Agency MBS Index Prepayment Model

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Key updates for BAM GNPL model V1.43.1

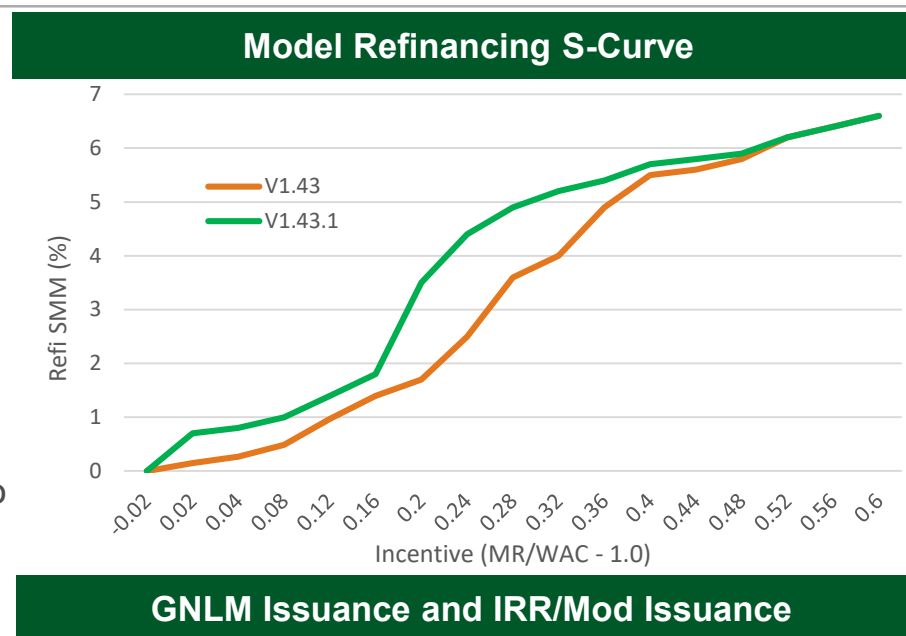
Summary of key model changes

Model Subcomponent	Changes
Rate/Term Refinancing	Steepened S-curve Flattened refi penalty curve Added refi aging multiplier for “Healthcare” vs. “Multifamily” Added refi aging multiplier for “New Construction” vs. “Refi/Purchase”
Cashout Refinancing	Increased cashout lock-in effect Flattened cashout penalty curve Added cashout aging multiplier for “Healthcare” vs. “Multifamily” Added cashout aging multiplier for “New Construction” vs. “Refi/Purchase”
Economic Incentive	Increased penalty IO multiplier to reduce penalty elbow shift

Rate/Term Refinance

Rate/term refis: steeper S-curve in V1.43.1

- Interest Rate Reduction (IRR) transactions have increased greatly in recent years.
- Prepayment penalties can be paid by lenders to reduce refinancing costs.
- HUD encourages rate refi or IRR, in order to improve properties' operating cash flows and reduce the chance of insurance claims.
- We have slightly elevated our model's S-curve to reflect increased IRR activity.

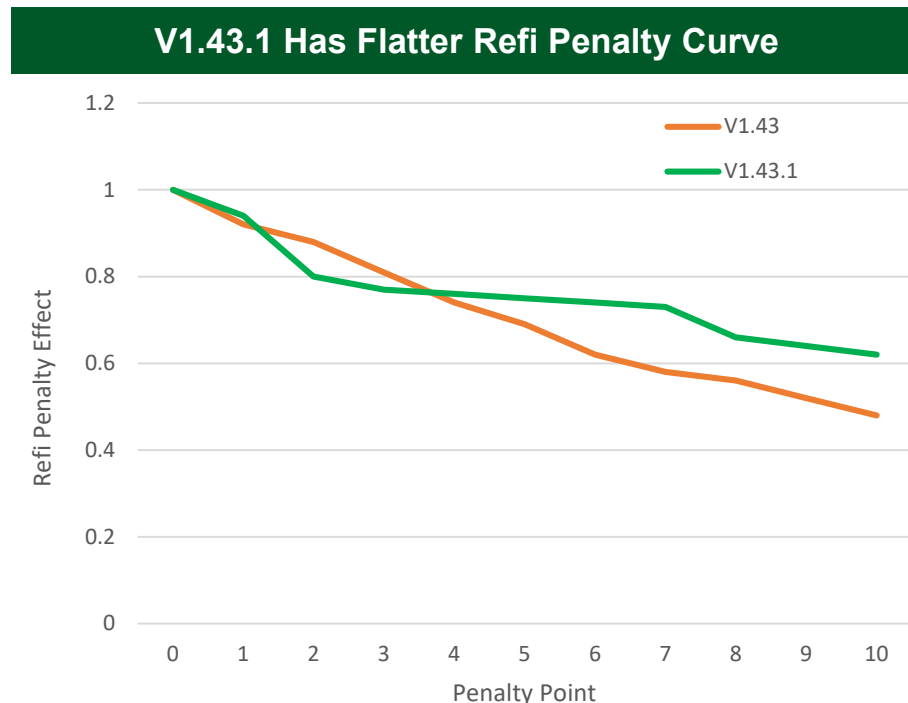


Year	Total Issuance (\$BN)	IRR/Mod Issuance (\$BN)	GNLM IRR/Mod %
2013	1.1	0.7	61.7
2014	3.1	2.8	88.6
2015	6.3	6.0	94.3
2016	4.6	4.3	93.2
2017	1.6	1.5	96.9
2018	0.2	0.2	100.0
2019	0.9	0.9	99.5
2020	11.7	11.7	99.8
2021	15.8	15.8	100.0
2022 (YTD)	1.9	1.9	100.0

Source: Bloomberg. As of April, 2022.

Rate/term refis: Rising IRR has weakened the penalty effect

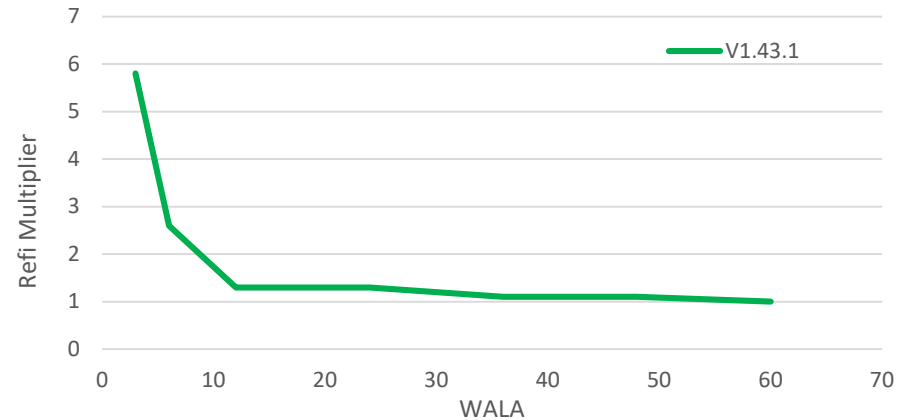
- In an IRR transaction, lenders can directly pay the penalties for borrowers, eroding the penalty effect.
- Other factors, such as penalty tax deductibility, whether the penalty amount can be rolled into the new loan, etc., can also affect a borrower's refinancing decision.
- As a result, we have flattened our model's refi penalty curve. In addition, we have also reduced the refi penalty elbow.



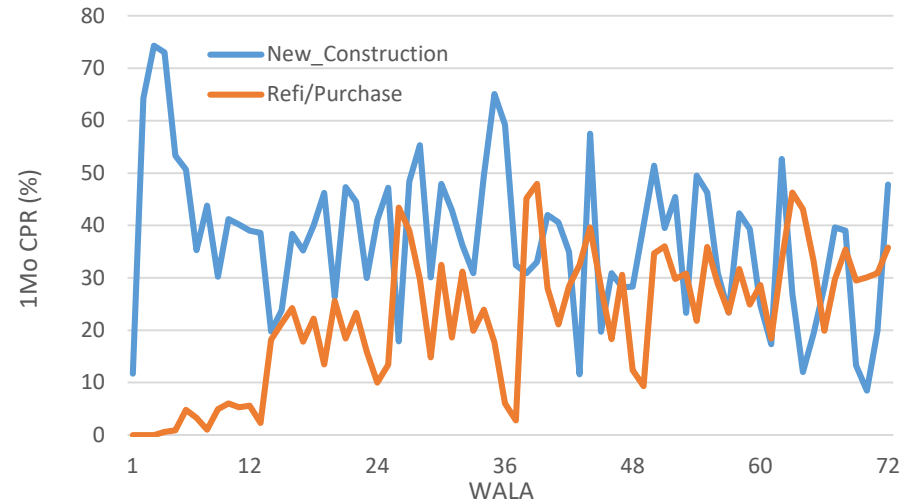
Rate/term refis: “New construction” loans prepay faster than “refi/purchase”

- During the construction phase, no principal can be paid back. So a construction loan is effectively in “lockout”.
- Right after conversion, “new construction” (NC) loans exhibit strong “pent-up” refi demand, compared with newly originated “refi/purchase” (RP) loans.
- We have added a refi multiplier in our model to account for the faster rate refi prepayments of “new construction” loans.

“NC” Refi Multipliers over “RP”



“New Construction” vs. “Refi/Purchase”

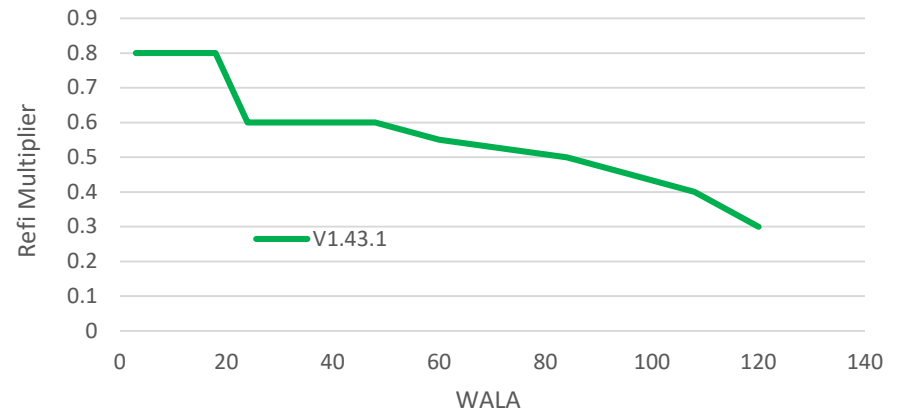


Source: Bloomberg. GNPN pool level data from Jan., 2020 to April, 2022.

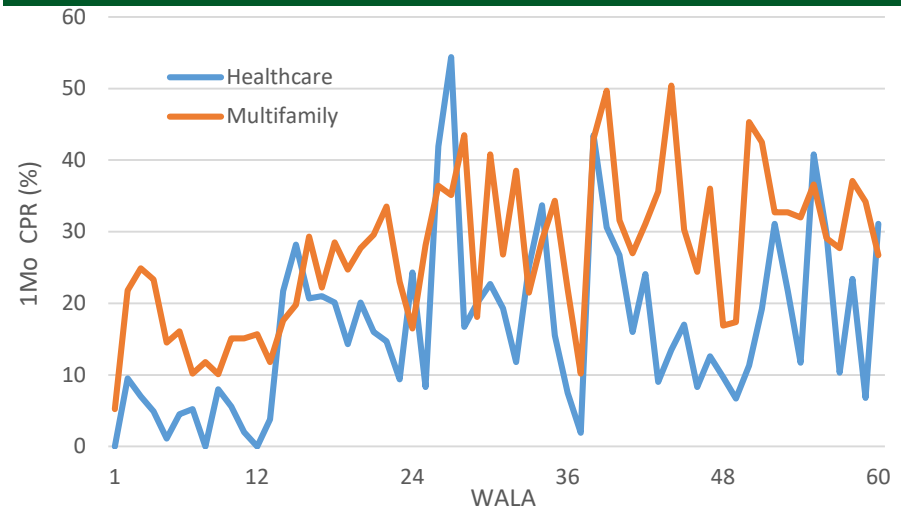
Rate/term refis: “Healthcare” loans prepay slower than “multifamily”

- Healthcare loans require higher DSCR and MIP than multifamily loans.
- Under similar rate incentives, healthcare loans prepay more slowly than multifamily loans.
- We have added a refi multiplier in our model to account for the slower prepayments of “healthcare” versus “multifamily” properties.

“Healthcare” Refi Multipliers over “Multifamily”



“Healthcare” vs. “Multifamily”

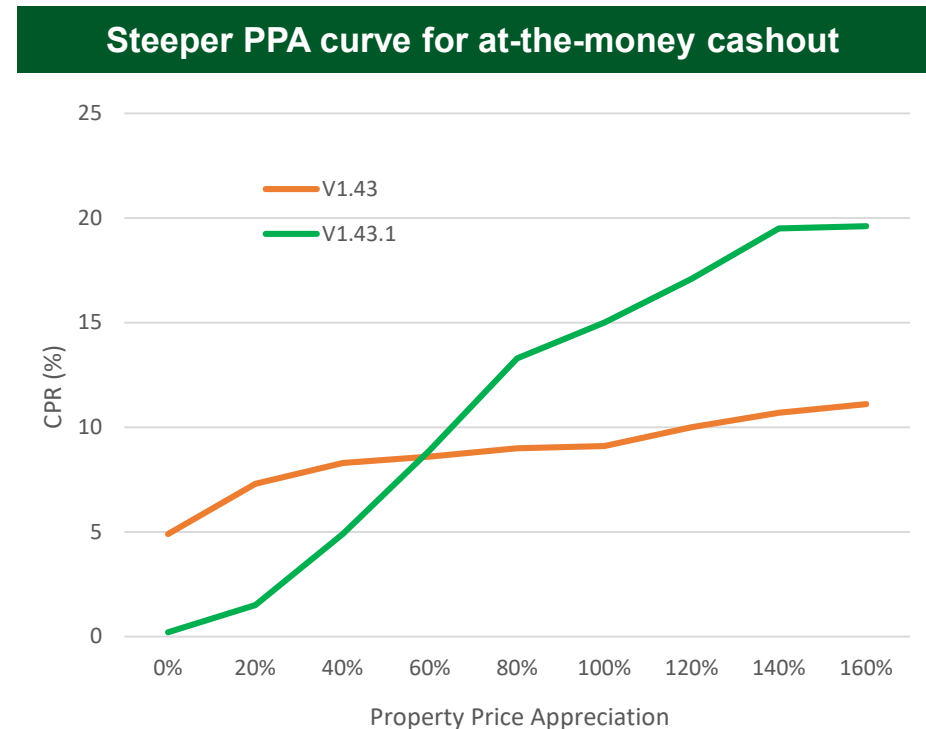


Source: Bloomberg. GNPN pool level data from Jan., 2020 to April, 2022.

Cashout

Cashout refis: V1.43.1 has a steeper property price appreciation (PPA) curve

- Property price appreciation is the primary driver of GNPL cashout refinancing.
- The cashout PPA curve of V1.43, the current production model, is too flat versus recent empirical data.
- We have steepened the cashout PPA curve as implied by the prepayment data of at-the-money GNPL pools.

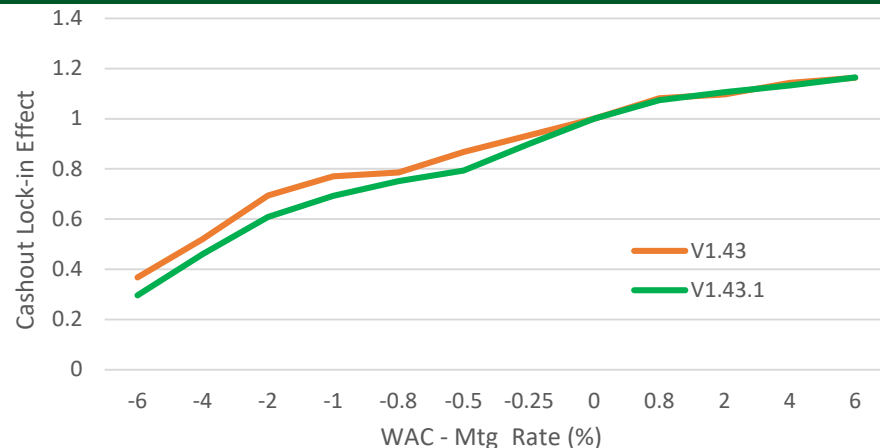


Source: Bloomberg. Projected prepayments for GNPN pools between Jan., 2015 and April, 2022, with corresponding RCA CPPI Apartment Price Index data.

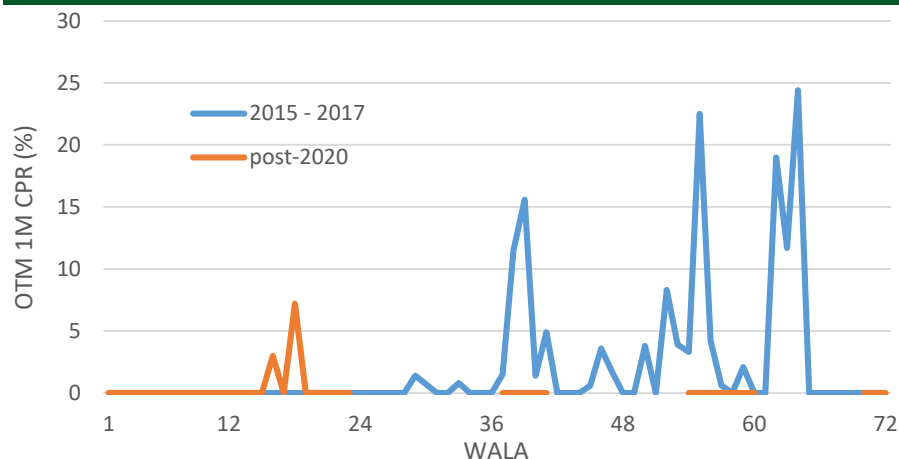
Cashout refis: OTM cashout refinancing projections have been lowered

- Although fundamentals for cashout refinancing have improved, HUD guidelines have put restrictions on cashout refinancing.
- As a result, since 2020 out-of-the-money (i.e., discount) aging curve seems to have lengthened
- We have adjusted our model's cashout lock-in effect and the discount cashout aging curve accordingly.

OTM Cashout lock-in is stronger in V1.43.1



Weaker OTM cashout after 2020

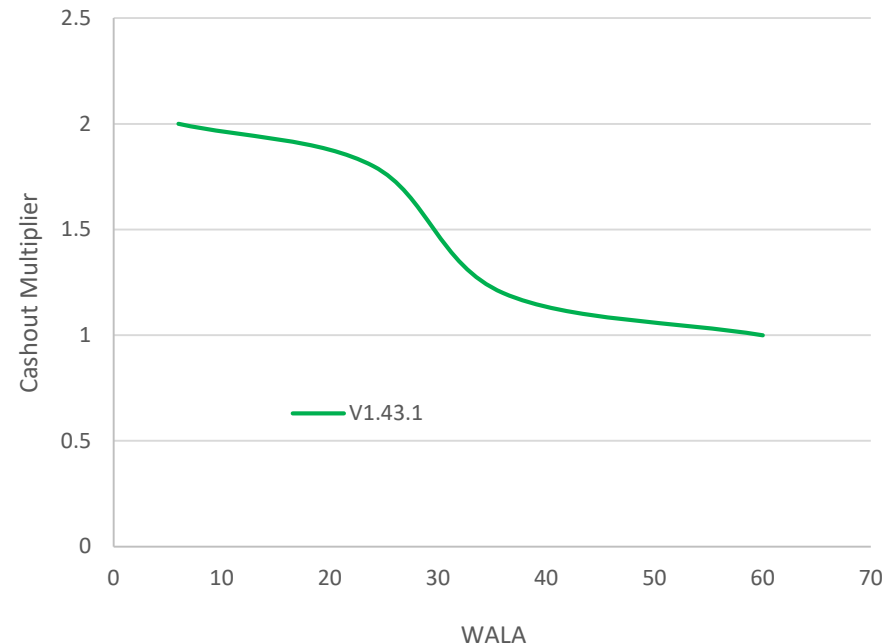


Source: Bloomberg. GNPN multifamily refi/purchase pools 50 to 100 bps OTM, between Jan., 2015 and April, 2022.

Cashout refis: “New construction” loans cashout more than “refi/purchase”

- Construction loans are usually sized based on cost. Once the construction is completed and the property is “leased-up”, its market value is usually higher than the original cost.
- Other favorable factors for cashout refinancing include:
 - Property prices have been going up since mid-2010.
 - Rates have generally been dropping since late 2018.
- We have added a cashout multiplier in our model to account for the faster cashout of new construction loans.

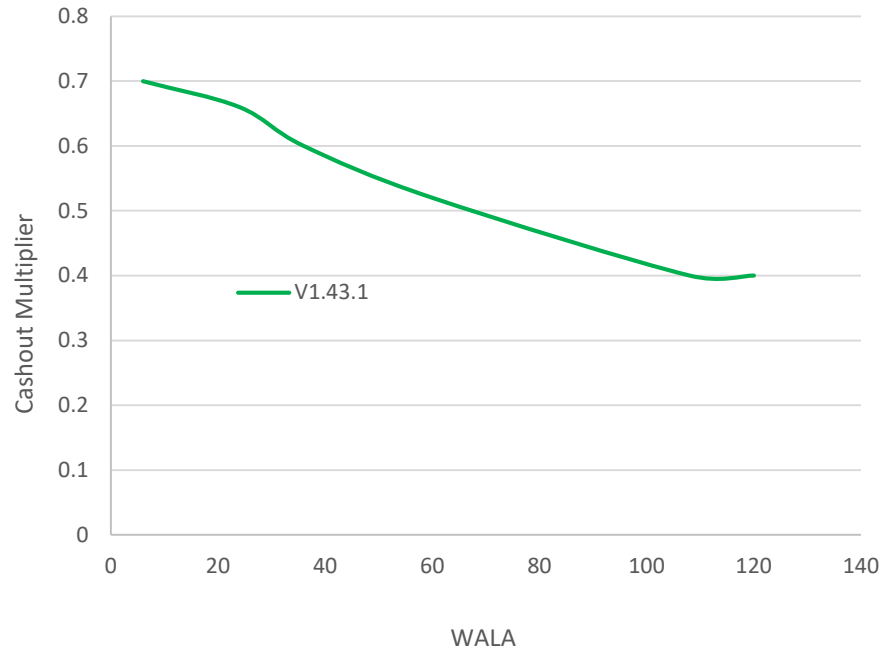
“NC” cashout Multipliers over “RP”



Cashout refis: “Healthcare” loans have less cashout than “multifamily”

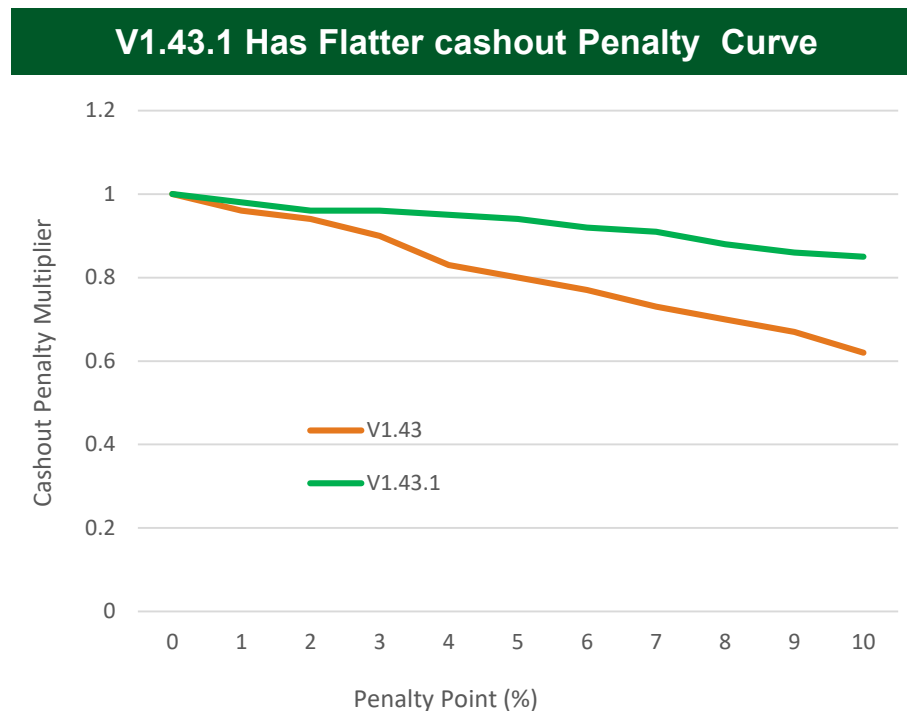
- Certain FHA-insured healthcare properties (e.g., nursing homes) have per-unit construction caps, with lower valuations than multifamily.
- Healthcare properties generally have lower property price appreciation due to fixed revenue streams, high labor costs, etc.
- We have added a cashout multiplier in our model to account for the lower cashout behavior of “healthcare” loans.

“Healthcare” cashout Multipliers over “Multifamily”



Cashout refis: Cashout sensitivity to penalties has weakened

- Although prepayment penalties are a deterrent for cashout refinancing, its effect has become weaker
- Borrowers determined to overcome HUD's restrictive cashout rules have been cashing out with little regard to prepayment penalties incurred
- We have flattened the model's cashout penalty curve to reflect the lower cashout sensitivity to prepayment penalties.



Projections and Analytics

Summary of GNPL projections and risk analytics (v1.43.1 vs. v1.43)

Impact on prepayment projections

- Except for deep discounts, 1-Yr projections are 0-10 CPR faster due to higher rate refinancing and cashouts.
- In the -200 MR rate scenario, prepayments increase by 7-23 CPR due to stronger refinancing.
- In the +200 MR rate scenario, prepayments increase by 0-8 CPR due to higher cashout activity.

Impact on risk analytics

- OADs shorten by 0.2-1.1 years across the board.
- OAS largely increases by 5-21bp, except for some seasoned coupons where they fall by 0-4bp.
- The resulting OAS profile is flat across the belly part of the coupon stack.

Comparison of model projections under various MR shift scenarios

1Y CPR projections for benchmark GNPL pools																	
			V1.43					V1.43.1					Change				
Cpn	Vtg	Bmk Pool	-200	-100	0	100	200	-200	-100	0	100	200	-200	-100	0	100	200
2	2021	GN BS9420	6.4	5.2	4.6	4.0	3.4	5.9	4.3	3.8	3.3	2.8	-0.5	-0.9	-0.9	-0.7	-0.6
	2020	GN BW4572	6.7	5.5	4.9	4.3	3.7	7.0	5.2	4.6	4.0	3.4	0.3	-0.2	-0.3	-0.2	-0.2
	2012	GN 749598	8.5	6.9	6.2	5.4	4.7	22.5	17.1	15.3	13.6	11.7	14.0	10.1	9.1	8.2	7.1
2.5	2021	GN BV7078	6.7	5.0	4.4	3.8	3.2	6.2	3.4	2.9	2.6	2.2	-0.4	-1.5	-1.5	-1.2	-1.0
	2020	GN CB0941	7.0	5.4	4.8	4.2	3.7	7.7	4.9	4.2	3.7	3.2	0.7	-0.4	-0.6	-0.6	-0.5
	2013	GN 768311	10.1	7.5	6.8	6.0	5.1	29.0	18.8	16.8	15.0	13.1	18.9	11.3	10.0	9.1	8.0
	2012	GN AC1397	9.6	6.8	6.2	5.4	4.7	25.3	15.9	14.2	12.7	11.1	15.6	9.1	8.0	7.3	6.4
3	2021	GN BY8342	7.4	6.2	5.0	4.5	3.9	16.2	5.6	4.3	3.8	3.3	8.8	-0.6	-0.8	-0.7	-0.6
	2020	GN BR3544	8.1	6.4	5.4	4.8	4.1	22.9	6.9	5.7	5.1	4.4	14.8	0.5	0.4	0.3	0.3
	2019	GN BE6954	11.7	6.6	5.4	4.8	4.2	24.6	7.9	6.1	5.4	4.7	12.9	1.3	0.7	0.6	0.6
	2018	GN AZ2600	22.7	7.2	5.8	5.1	4.5	33.1	13.7	9.3	8.2	7.2	10.5	6.5	3.6	3.1	2.8
	2017	GN AY8873	17.2	6.9	5.7	5.1	4.4	32.2	14.5	10.7	9.5	8.3	15.0	7.6	5.1	4.4	3.9
	2016	GN AU4037	18.0	6.7	5.5	4.9	4.3	31.7	14.3	10.1	8.9	7.8	13.8	7.6	4.6	4.0	3.6
	2015	GN AP6685	23.1	7.0	5.6	5.0	4.4	39.6	17.6	11.3	10.0	8.8	16.5	10.6	5.7	5.0	4.4
3.5	2020	GN BR3560	21.8	7.3	5.6	5.0	4.4	45.2	9.3	6.0	5.3	4.7	23.4	1.9	0.4	0.3	0.3
	2019	GN BN1936	20.7	7.0	5.3	4.8	4.2	37.3	9.9	6.5	5.7	5.1	16.5	2.9	1.2	1.0	0.9
	2018	GN BG1344	38.0	7.7	5.9	5.3	4.6	46.9	16.2	9.6	8.1	7.2	8.9	8.5	3.7	2.8	2.5
	2017	GN AU1230	34.5	7.7	6.0	5.3	4.7	48.1	23.4	12.2	10.8	9.6	13.6	15.7	6.2	5.5	4.9
	2016	GN 773590	39.7	7.8	6.0	5.4	4.7	47.1	25.2	12.8	10.8	9.6	7.4	17.4	6.8	5.5	4.9
	2015	GN AB8592	36.5	7.8	6.2	5.5	4.8	48.2	23.3	13.7	12.1	10.7	11.7	15.6	7.5	6.6	5.9
	2014	GN AD5912	43.3	10.9	6.8	6.0	5.3	51.0	33.0	17.1	14.3	12.7	7.7	22.1	10.4	8.3	7.4
4	2019	GN BM3660	26.7	7.5	6.1	5.1	4.5	47.4	11.7	8.1	6.6	5.8	20.7	4.2	2.0	1.5	1.3
	2018	GN BI8572	43.5	8.1	6.4	5.4	4.8	53.8	18.9	9.1	7.5	6.6	10.3	10.8	2.7	2.0	1.8
4.5	2018	GN BG4531	43.9	14.7	7.3	5.8	5.1	50.9	28.1	14.0	9.3	8.0	7.0	13.4	6.7	3.5	2.9

Source: Bloomberg. As of 04/29/2022 market close, for selected multifamily refi/purchase GN project loan pools.

Comparison of model valuations and risk metrics

BAM analytics for benchmark GNPL pools (V1.43.1 vs. V1.43)

Cpn	Vtg	Bmk Pool	V1.43						V1.43.1					Change				
			Price	Yield	N-sprd	LOAS	LOAD	LOAC	Yield	N-sprd	LOAS	LOAD	LOAC	Yield	N-sprd	LOAS	LOAD	LOAC
2	2021	GN BS9420	89.8945	3.5	54.7	23.1	8.0	0.1	3.6	63.9	34.1	7.6	0.1	0.1	9.2	11.0	-0.4	0.0
	2020	GN BW4572	90.3672	3.5	48.3	15.8	7.8	0.1	3.6	62.2	31.4	7.4	0.1	0.1	14.0	15.5	-0.5	0.0
	2012	GN 749598	99.5508	2.2	-84.1	-114.5	6.2	-0.5	2.2	-83.6	-118.7	5.1	-0.7	0.0	0.5	-4.3	-1.1	-0.2
2.5	2021	GN BV7078	93.4961	3.5	52.5	15.3	7.7	-0.1	3.5	53.7	19.2	7.4	-0.2	0.0	1.2	3.9	-0.2	-0.1
	2020	GN CB0941	94.5039	3.5	51.7	12.4	7.4	-0.2	3.6	57.7	21.4	7.1	-0.2	0.1	6.1	9.0	-0.4	0.0
	2013	GN 768311	98.8438	2.5	-48.2	-86.8	5.8	-0.8	2.6	-44.1	-84.2	4.7	-0.8	0.1	4.1	2.6	-1.1	0.0
	2012	GN AC1397	99.1055	2.4	-59.5	-92.9	6.2	-0.7	2.4	-57.4	-94.0	5.3	-0.7	0.0	2.2	-1.1	-0.9	0.0
3	2021	GN BY8342	97.5469	3.6	59.6	14.1	6.9	-0.5	3.6	62.2	18.9	6.6	-0.4	0.0	2.7	4.8	-0.3	0.1
	2020	GN BR3544	96.9219	3.5	51.2	4.9	6.7	-0.6	3.6	59.5	16.2	6.3	-0.5	0.1	8.2	11.3	-0.4	0.1
	2019	GN BE6954	97.1094	3.5	49.7	1.8	6.7	-0.6	3.6	59.3	15.6	6.2	-0.5	0.1	9.7	13.8	-0.4	0.1
	2018	GN AZ2600	97.7539	3.4	43.4	-8.4	6.1	-0.9	3.6	58.8	12.8	5.5	-0.5	0.2	15.4	21.2	-0.7	0.4
	2017	GN AY8873	97.6406	3.4	42.8	-5.8	6.2	-0.8	3.6	59.5	14.6	5.3	-0.6	0.2	16.7	20.4	-0.8	0.2
	2016	GN AU4037	97.1250	3.4	39.2	-7.2	6.3	-0.8	3.5	54.0	8.6	5.5	-0.6	0.2	14.8	15.8	-0.8	0.2
	2015	GN AP6685	97.6641	3.5	51.8	4.9	6.1	-0.9	3.6	64.0	16.7	5.3	-0.7	0.1	12.2	11.8	-0.8	0.2
3.5	2020	GN BR3560	100.1602	3.6	59.3	0.2	6.3	-0.8	3.6	63.4	8.3	5.9	-0.7	0.0	4.1	8.2	-0.4	0.2
	2019	GN BN1936	99.9805	3.6	56.6	2.4	6.2	-0.8	3.6	62.7	12.9	5.8	-0.6	0.1	6.1	10.5	-0.4	0.2
	2018	GN BG1344	101.0430	3.6	61.8	0.9	5.5	-1.0	3.7	68.7	13.6	5.0	-0.7	0.1	6.8	12.7	-0.5	0.3
	2017	GN AU1230	98.9219	3.6	57.9	2.5	5.7	-1.0	3.7	71.7	22.0	5.0	-0.7	0.2	13.8	19.5	-0.7	0.3
	2016	GN 773590	99.3672	3.7	71.0	15.6	5.6	-0.9	3.8	81.7	28.4	4.8	-0.8	0.1	10.8	12.9	-0.8	0.1
	2015	GN AB8592	97.9414	3.6	60.4	9.9	5.7	-1.0	3.8	73.8	24.9	4.8	-0.7	0.1	13.4	14.9	-0.9	0.3
	2014	GN AD5912	99.3984	3.8	77.1	24.6	5.2	-1.1	3.8	82.8	24.2	4.1	-1.1	0.1	5.7	-0.3	-1.1	0.0
4	2019	GN BM3660	99.8945	4.1	109.9	40.4	5.5	-0.6	4.2	117.5	58.0	5.3	-0.3	0.1	7.6	17.6	-0.2	0.2
	2018	GN BI8572	102.6484	3.7	68.3	1.4	5.3	-0.9	3.7	70.6	8.2	4.8	-0.8	0.0	2.3	6.7	-0.5	0.1
4.5	2018	GN BG4531	104.6211	3.8	80.2	-0.9	4.8	-1.5	3.8	76.8	-1.5	4.1	-1.2	0.0	-3.4	-0.5	-0.7	0.2

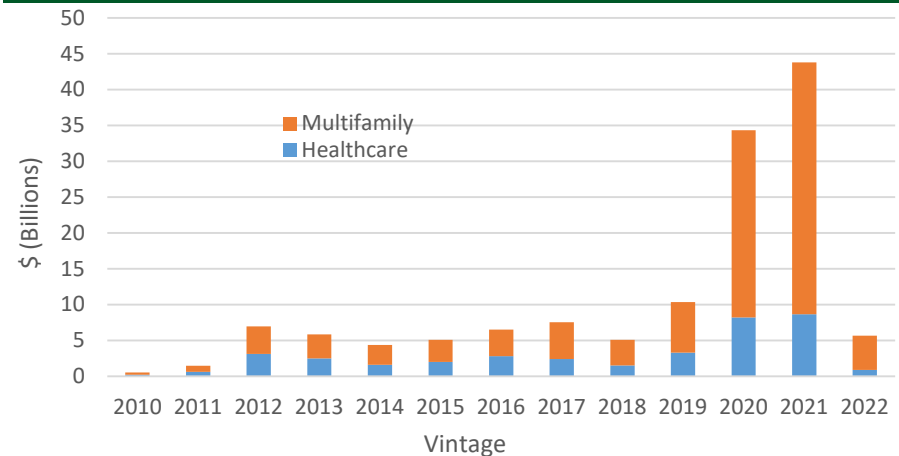
Source: Bloomberg. As of 04/29/2022 market close, for selected multifamily refi/purchase GN project loan pools.

Overview of the GN Project Loan Sector

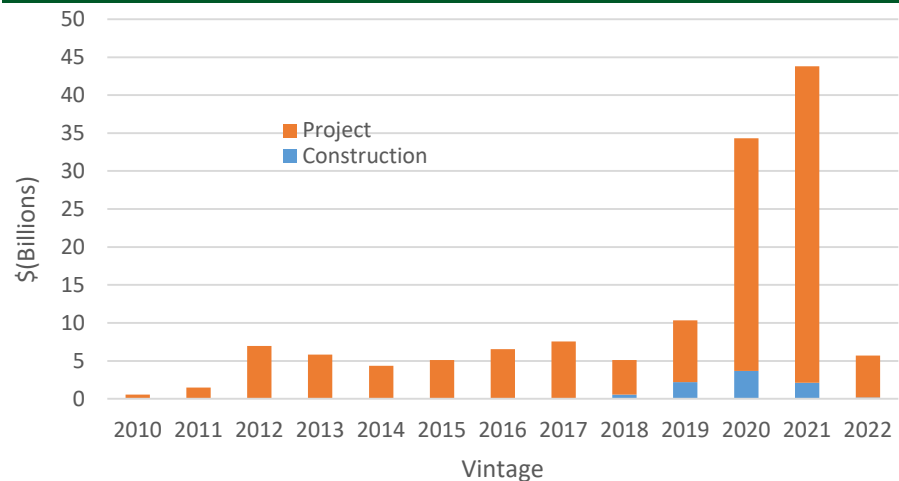
What are GN project loans?

- GN project loans are government (e.g., FHA) insured commercial property loans to finance multifamily housing (MF) or healthcare/nursing facilities (HC).
- GNPL pools carry explicit government guarantees of principal and interest payment by Ginnie Mae.
- Many GNPL pools, which include both construction and permanent project loans, back GNR CMO deals.
- Construction loans normally convert to permanent project loans upon the completion of construction.

Current Balance by Property Type



Current Balance by Loan Type

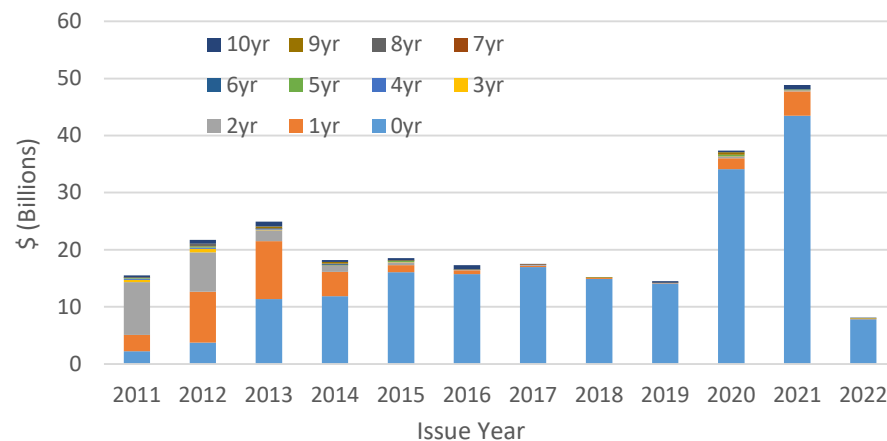


Source: Bloomberg. As of April, 2022.

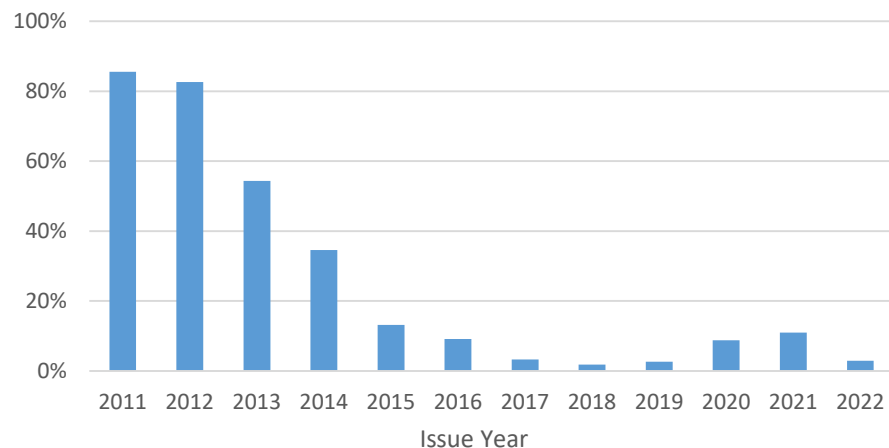
GN project loans often have a prepayment lockout period

- Unlike single family loans, GN project loans carry prepayment restrictions such as prepayment lockouts and/or penalties.
- Typically, a GN project loan may start out with a lockout period (where borrowers are not allowed to refinance), followed by a penalty period.
- The combined length of lockout and penalty period is normally 10 years.
- Over the years, lockout terms have become shorter. Since 2011, most of the new originations have short (e.g., 1 or 2 years) or no lockout terms at all.

Project loan issuance by lockout term



Percent of issuance with lockout

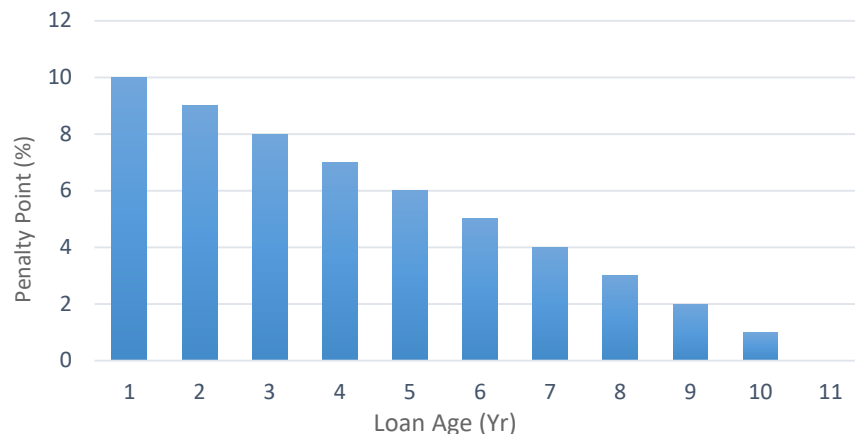


Source: Bloomberg. As of April, 2022.

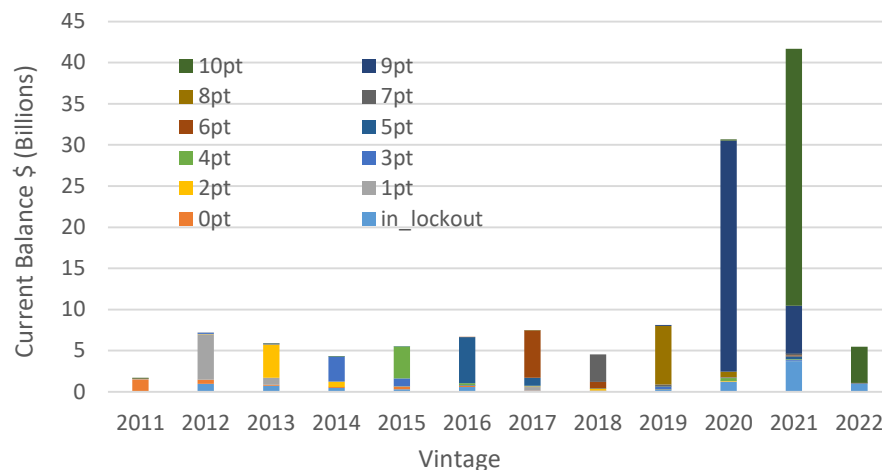
Most GN project loans have prepayment penalties

- Following the lockout period, the starting penalty rate is typically equal to 10 minus the lockout term.
- The penalty typically declines 1% annually (“step-down”) until it reaches 0% after year 10.
- Without lockout, penalty rates typically have the pattern of “10, 9, 8, 7, ..., 0”. Recently, the structure of “10, 10, 10, 7, ..., 0” has gained some popularity.
- Relative to other types of prepayment penalties, “step-down” is a relatively weak form of call protection.

Typical GNPL penalty structure by loan age



Project loan penalty by vintage

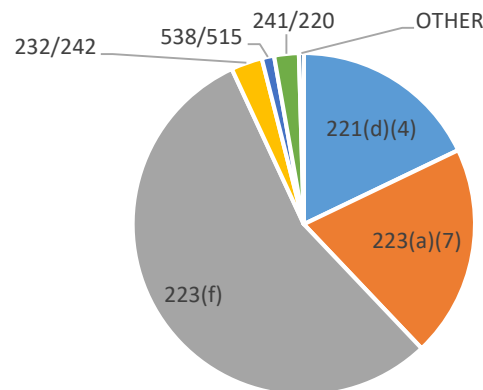


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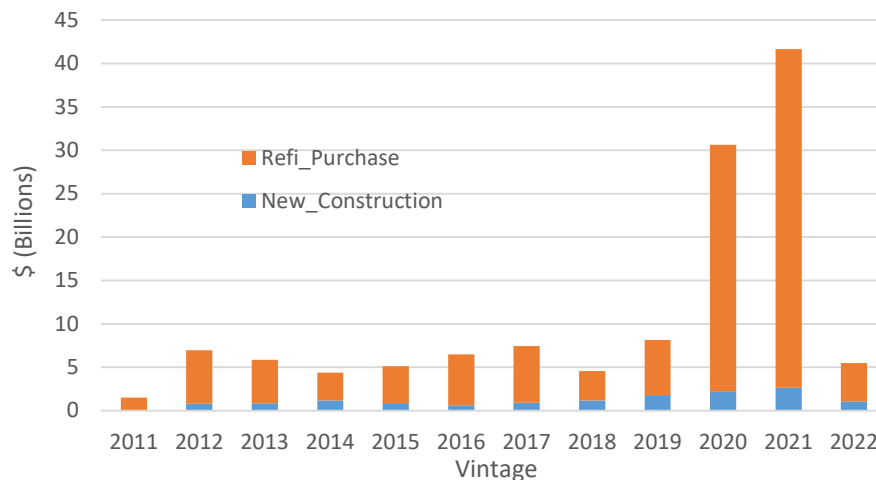
FHA program code and loan purpose

- Most GN project loans are insured by the FHA under various sections of the National Housing Act (aka FHA program code).
- The most common FHA multifamily program codes are listed below:
 - 223(f): acquisition (purchase or refinance) of (mostly) non-FHA insured properties. cashout is allowed.
 - 223(a)(7): streamlined refinancing for existing FHA-insured loans. cashout is not permitted.
 - 221(d)(4): multifamily for-profit new construction or substantial rehabilitation loans.
 - 232/242: healthcare/senior/hospital new construction loans.
 - 241/220: multifamily improvement or rehabilitation/construction.
 - 538/515: USDA extreme low income multifamily loans.

Current balance distribution by FHA program



Current balance by loan purpose

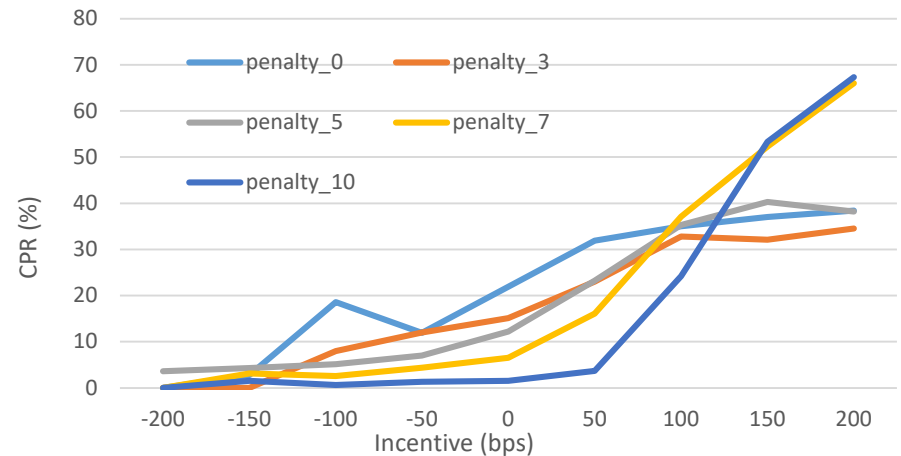


Source: Bloomberg. As of April, 2022.

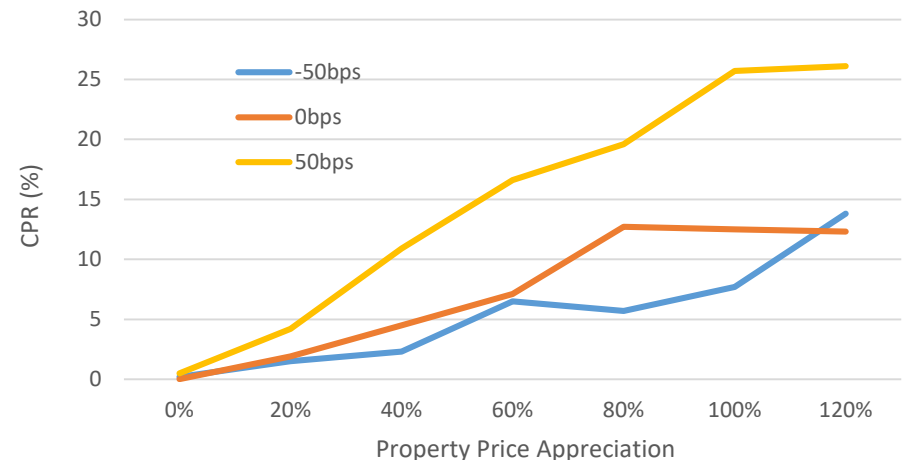
Prepayment sensitivity to mortgage rates and property prices

- In general, a project loan's refinancing sensitivity is effectively dampened by prepayment penalties.
- However, once rate incentives become high enough (e.g., above 100 ~ 150 bps), the prepayment penalty effect diminishes.
- Cashout refinancing, which is largely driven by property price appreciation (PPA), plays an important role in project loan prepayments.

Project loan S-curve tiering by penalty point



Project loan speeds PPA curve by rate incentive

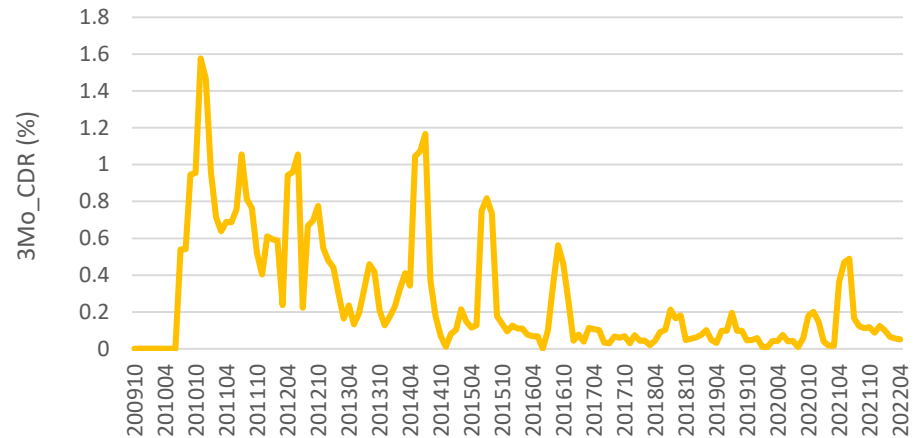


Source: Bloomberg. GNPN pool data between Jan., 2015 and April, 2022 with RCA CPPI Apartment Price Index.

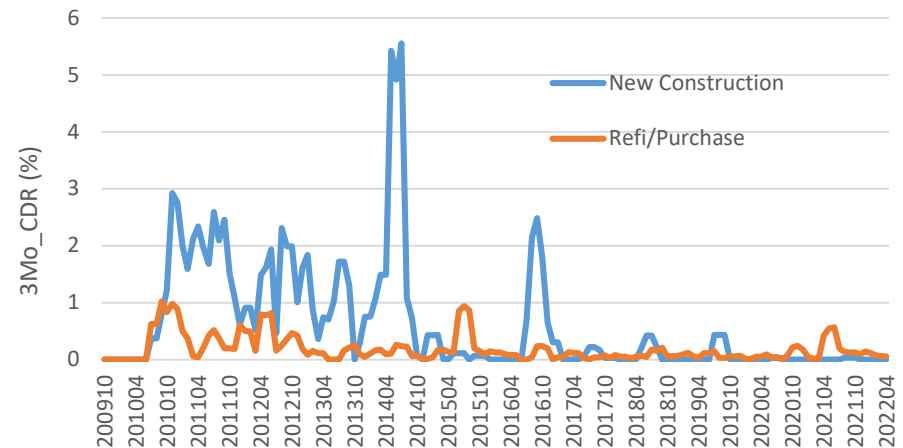
Defaults are typically low and stable

- The overall project loan CDR level has been mostly below 1% and stable since the financial crisis of 2008.
- However, new construction loans can have noticeably higher delinquency/buyout rates than their refi/purchase counterparts ...
- ... as they must face higher uncertainties (e.g., “lease-up” risk) during the “stabilization” period.

Project loan overall CDR



Project loan CDR by loan purpose



Source: Bloomberg. GN project loan pool data between Oct., 2009 and April, 2022

BAM GN Project Loan Model Structure

BAM GN project loan model overview

- BAM projects the prepayment behavior of GN project loans/pools
- There is one main “base” model for the commonalities of all loan types.
 - Project Loans (lockout, prepayment penalties, multifamily or healthcare, 221(d)(4), 223(f), 232, 223(a)(7), etc.)
 - Construction Loans (both CL and CS, buyout only)
- Pool types are grouped and modelled as three sub types
 - PN, PL and LS: the most common types of multifamily and healthcare properties
 - LM: recently mostly “Interest Rate Reduction” loans; also for seasoned or modified loans
 - RX: mark-to-market of rental-assistance properties
- The model produces **two** vectors of prepayment rates (VPRs, CDRs) based on
 - Collateral characteristics (lockout and prepayment penalty, FHA program, loan size, etc.)
 - Market environment (mortgage rates, property prices, etc.)

Key components of overall GNPL prepayments

- Broadly, we can think of overall project loan prepayments as coming from three subcomponents

$$SMM_{Total} = SMM_{Refi} + SMM_{Cashout} + SMM_{Invol}$$

- Voluntary Prepayments
 - Rate/Term Refinancing: SMM_{Refi}
 - Cashout Refinancing: $SMM_{Cashout}$
- Involuntary Prepayments
 - Delinquency Buyouts: SMM_{Invol}
- Note: **SMM** (single monthly mortality rate) is the monthly prepayment rate on a pool of mortgages
CPR (constant prepayment rate) is the annualized prepayment rate on a pool of mortgages

Rate/Term refinancing

Factors that affect the rate refinancing profile

Collateral attributes

- Lockout
- Prepayment Penalty
- Loan Purpose (new construction, refi/purchase)
- Property Type (multifamily, healthcare)
- FHA Program (221(d)(4), 223(f), 232, 223(a)(7))
- SATO
- Loan Size
- Loan age (refi ramp)

Market environment

❖ Economic incentive to refinance

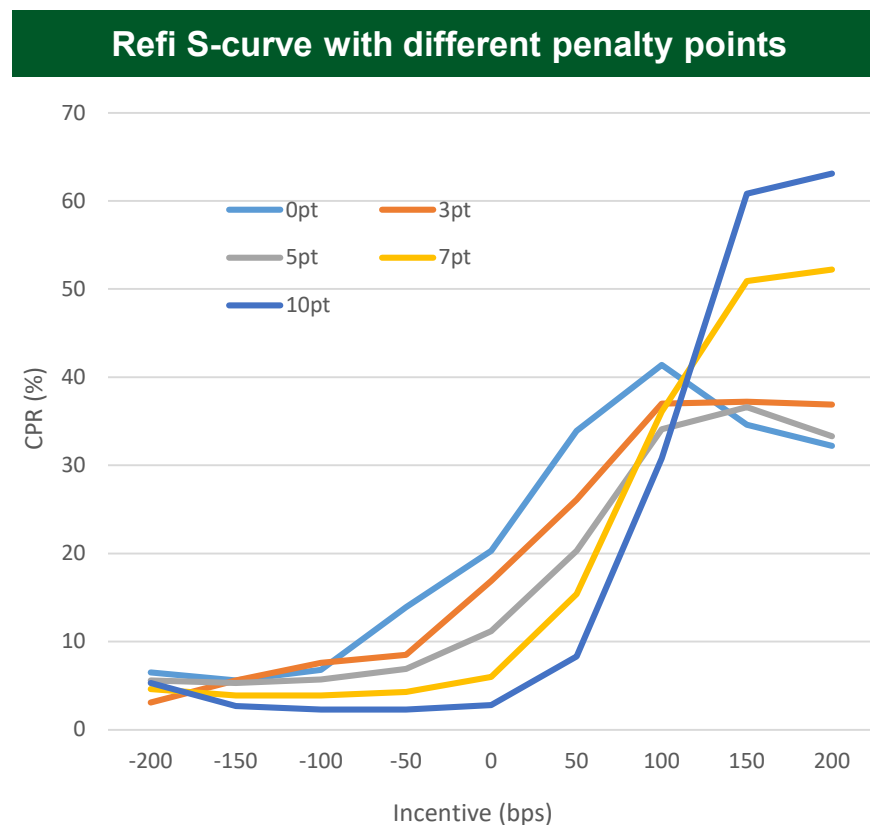
- Cumulative refi opportunities (burnout)
- Catch-up on refi opportunities immediately after lockout (pent-up effect)
- Property prices (updated LTV)

Underwriting standards / costs

- Prepayment penalty amount may be rolled into the new loan under certain programs (e.g., 223(a)(7), etc.).
- Prepayment penalty amount may even be paid by lenders under certain conditions (e.g., Interest Rate Reduction, etc.).

Mortgage rates, prepayment penalties, and refinancing

- While not in lockout, falling mortgage rates induce borrowers to refinance.
- Available rate incentives and existing penalties interact to determine a loan's refinancing behavior.
- Penalties can be an effective prepayment deterrent when rate incentives are moderate.
- BAM uses elbow shifts and multipliers to account for the impact of penalties on a loan's refinancing behavior.

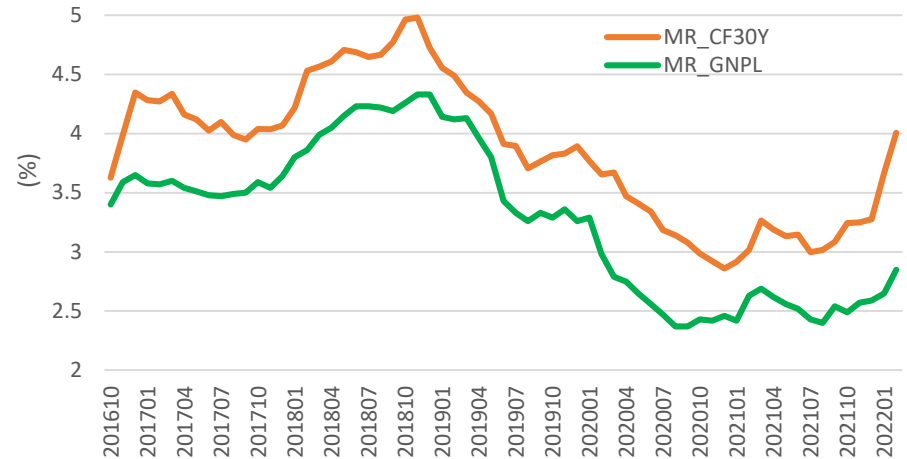


Source: Bloomberg. Projected prepayments for GNPN pools between Jan., 2015 and April, 2022.

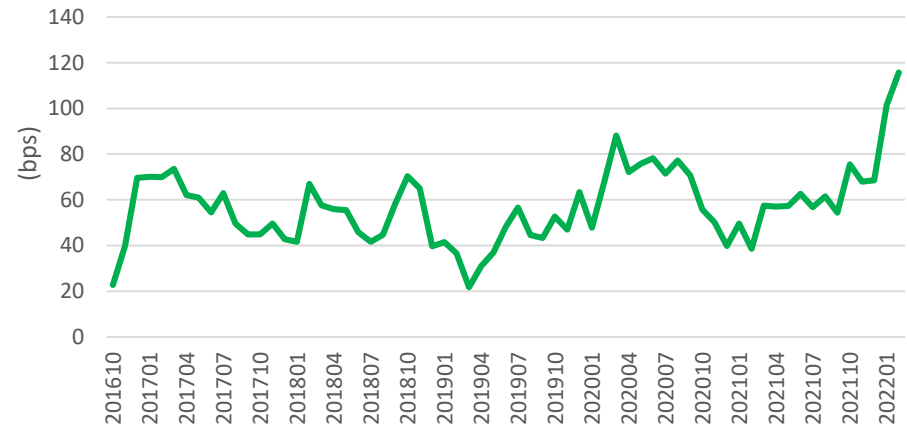
GNPL mortgage rate is ~50 bps lower than 30-yr conventional rate

- The current forward projection for GN project loan mortgage rates is about 50 bps below that of 30-yr conventional rate.
- Historically, GN project loan note rate has closely tracked the single family 30-yr conventional mortgage rate.
- The spread between the two rates has been stable and around 50 bps in recent years.

Mortgage Rates: Conventional 30Y vs. GNPL



Spread: Conventional 30Y over GNPL



Sources: Bloomberg, HUD, Ginnie Mae

Cashout refinancing

Factors that affect cashout refinancing

Collateral attributes

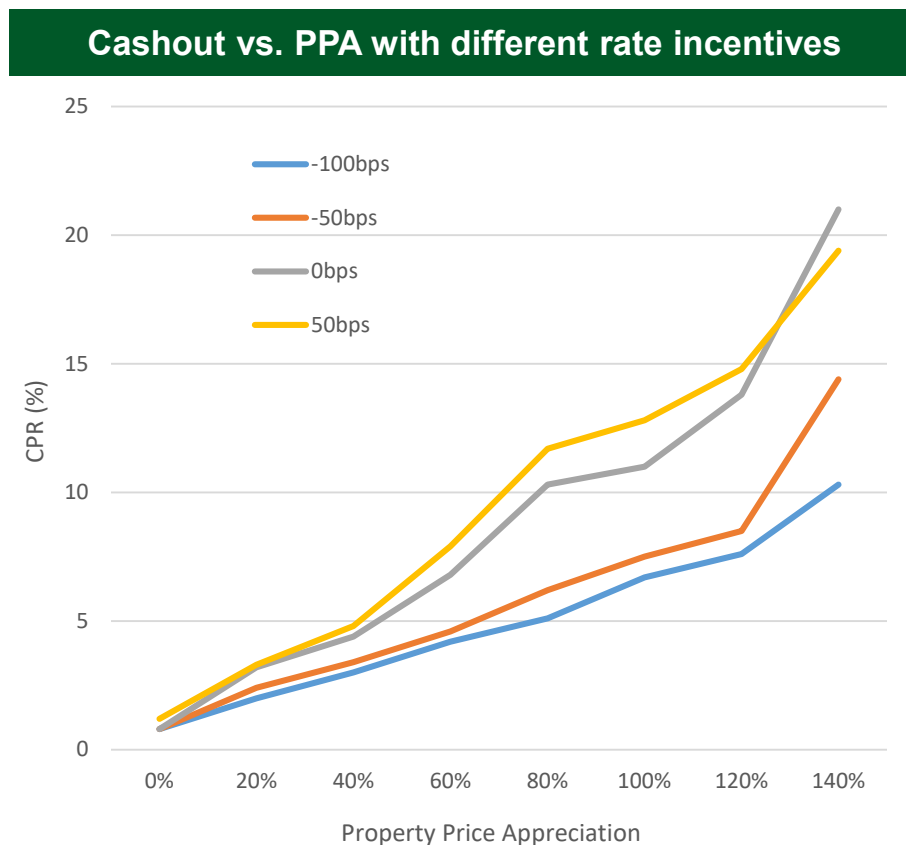
- Lockout
- Prepayment Penalty
- Loan Purpose (new construction, refi/purchase)
- Property Type (multifamily, healthcare)
- FHA Program (221(d)(4), 223(f), 232, 223(a)(7))
- 223(f) allows limited cashout
- 223(a)(7) or IRR doesn't allow cashout
- Loan Size

Market and regulatory environment

- ❖ Cumulative property price appreciation
- Cumulative cashout opportunities (burnout)
- Level of mortgage rates (lock-in)
- Catch-up on cashout opportunities immediately after lockout (pent-up effect)
- HUD restrictions on cashouts to reduce claims against the FHA insurance fund.

Cashout refinancing captures equity extraction activity

- As equity in the property increases, borrowers become increasingly likely to do a cashout refinance.
- Multifamily property prices have been increasing since 2010.
- Commercial property owners are more alert to opportunities to take out equity and leverage up, than single family home owners.
- Higher mortgage rates present an opportunity cost and discourage cashout activity (lock-in effect).



Source: Bloomberg. Projected prepayments for GNPN pools between Jan., 2015 and April, 2022, with RCA CPPI Apartment Price Index.

Delinquency buyouts

Factors that affect buyouts

Collateral attributes

❖ Loan age

- Lockout (no refi opportunities)
- Loan Purpose (new construction, refi/purchase)
- FHA Program (221(d)(4), 223(f), 232, 223(a)(7))
- SATO

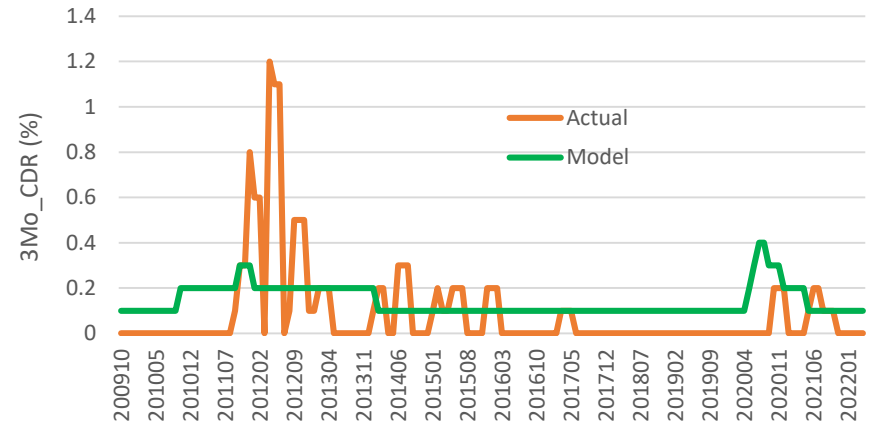
Market environment

- Property prices (cumulative PPA, updated LTV)
- Cumulative refi opportunities (burnout)
- Unemployment rate (rent payments)
- Mortgage rate (debt burden)

Loan age and loan purpose are key drivers for delinquencies

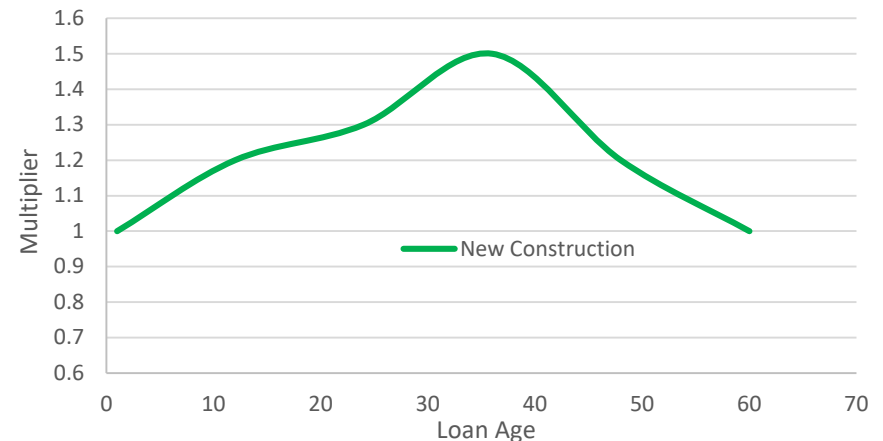
- In recent years, GNPL average default rates have generally been stable and below 1 CDR.
- Due to the CARES Act, even the Covid-19 pandemic hasn't caused default rates to rise substantially.
- All else equal, newly completed projects are more likely to face adverse operating cash flow situations (e.g., high vacancy rate, etc.) than established projects.

GNPL CDRs have been low and stable



Source: Bloomberg. Multifamily refi/purchase loans of vintage 2009 and later.

Boost for “New Construction” seasoning ramp



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